

Project 2 Report for ME 347 Spring 2026
Grading based on LO 1.5, 1.6, 1.7, 2.2, 3.1, 3.2, 3.3

Author Name: Sarah Tise
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1. Introduction

The goal of the device discussed in this document is to be a mast arm traffic light. This project presents the design and structural optimization of a commemorative traffic signal mast arm, commissioned to celebrate the centennial of the 1933 Chicago World's Fair. Inspired by the geometric simplicity of the cube, the proposed design utilizes a square-profile cantilever assembly to provide both aesthetic distinction and structural efficiency. Tasked with spanning a 26-foot roadway while accommodating multiple luminaire and signage loads, the assembly was rigorously evaluated against Chicago's unique environmental constraints, such as high winds. Through a comprehensive design iteration process utilizing SolidWorks for CAD modeling and ANSYS for Finite Element Analysis (FEA), this project balances municipal budgetary requirements with high-performance engineering, ensuring a structure that is both economically viable and capable of maintaining a factor of safety greater than 2.0 under all specified load cases.

2. Background and Design Objectives

When designing a traffic signal mast arm assembly, there are many mast arm designs to consider. Five of these designs are detailed in this section. The first mast arm design to consider is the standard tapered monotube.

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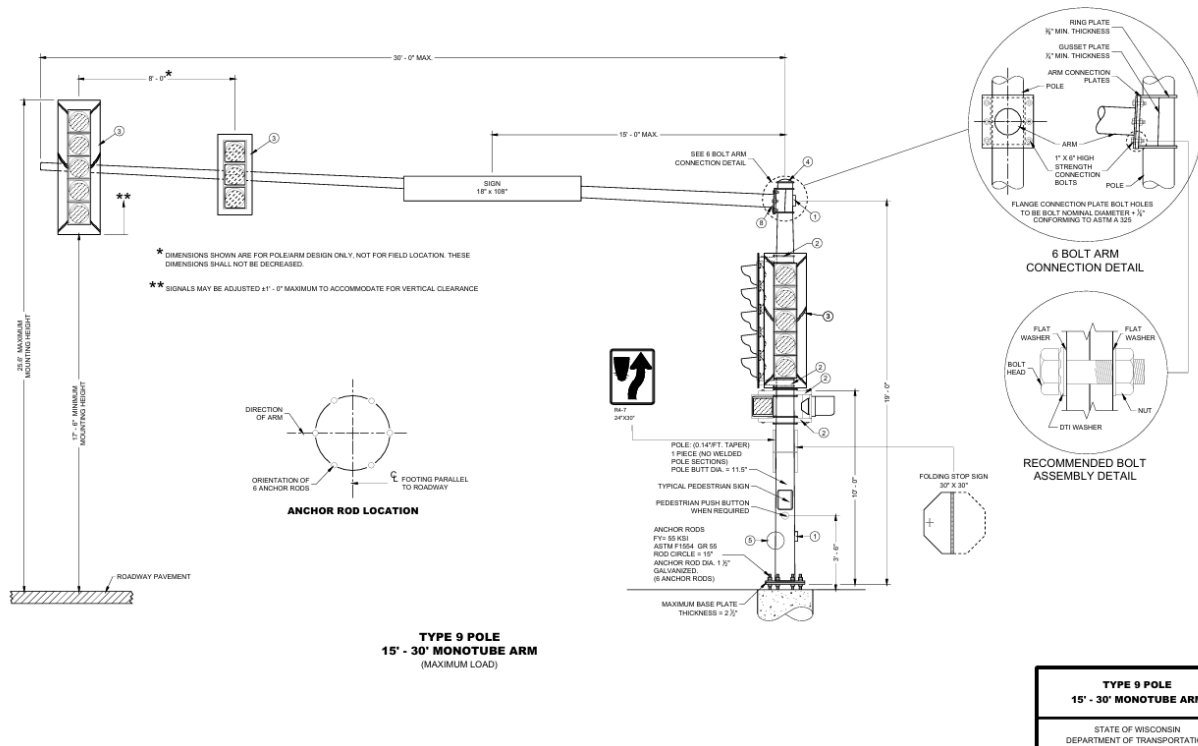


Figure 1: Standard Tapered Monotube Mast Arm Design, Type 9 Pole, Department of Transportation, Wisconsin

The [drawing](#) [1] above illustrates the distinguishing details of a standard monotube mast arm design. Note that the vertical pole tapers as the height increases as shown in the figure below. As shown in the drawing, the pole narrows by 0.14 inches per foot. The mast arm also clearly tapers towards its end.

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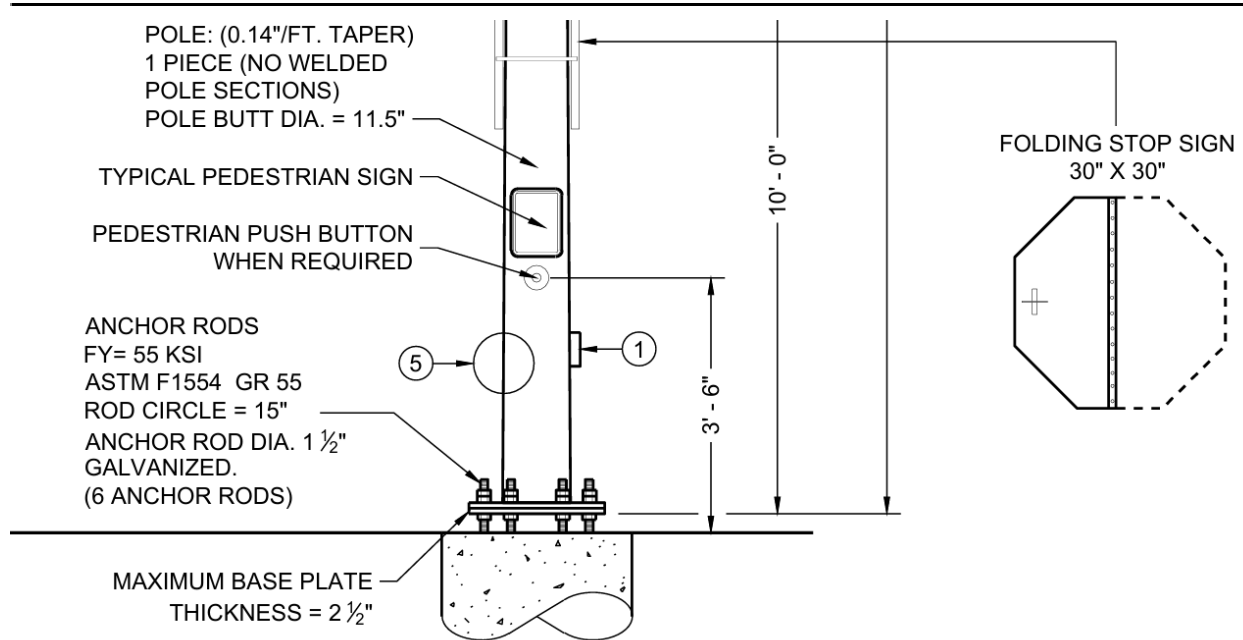


Figure 2: Standard Tapered Monotube Mast Arm Design, Taper Feature Highlighted

The next feature worth mentioning in this type of mast arm design is the monotube. Note how there is a single, continuously hollow tube for both the pole and the mast arm. This type is also often made of galvanized steel or high-strength aluminum.

As for how this design is likely to behave in weather often experienced in Chicago, this design is excellent for high winds since it is typically a round shape, and given the typical materials this type is made out of, this design would do well to withstand humidity and road salt.

The second design discussed in this document is the decorative mast arm, often seen in historical or ornamental contexts. The figure below illustrates an example of how mast arms can be designed to have more artistic flair. The image on the left [2] is an example of how the pole can be designed to have artistic ridges and bends purely for aesthetic purposes. The image on the right [3] is an example of how the mast arm can have bends or uniquely shaped ends that are purely for aesthetic purposes.

Given that this design often calls for the material to be cast aluminum or powder-coated steel, the extreme temperature swings that occur frequently in Chicago would likely lead to local corrosion or chipping of the coating, which would call for additional maintenance in order to prevent or repair this.



Figure 3: Decorative Pole Examples

The third design discussed in this document is the truss cantilever type. The image below [4] illustrates a commonly seen example of the truss cantilever mast arm design. The simple truss design can be seen in the mast arm, as shown below. A bottom and top chord are present, along with bracing between these chords, which creates a simple truss assembly. These are typically made out of structural steel or an aluminum alloy.

While the material options for this design don't raise any red flags for weather-induced corrosion, The truss design gives this mast arm design type a larger surface area compared to the other options, and so this design would carry a larger static load of ice, given the weather conditions of Chicago winters.



Figure 4: Truss Cantilever Mast Arm Example

The fourth mast arm type is the twin truss design. This is similar to the previous truss cantilever mast arm type discussed, however it varies slightly due to featured twists or turns seen in the truss assembly. These twists and turns allow signs, lights, etc, to be hung at different heights as shown in the images [5],[6] below. The angled mast arm allows the structure to better resist high winds when compared to the simple truss cantilever design. This type is typically manufactured out of structural steel.

[illegible]

The fifth mast type is the gusseted box beam arm. This design features a square cross-section rather than a circular one, as shown in Figure 7 [7]. The pole-to-arm interface is often reinforced with gussets, as shown in Figure 8 [8].

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This type is typically manufactured out of corten steel or painted structural steel. Due to the square cross-section, moisture can easily be trapped in the hollow areas if proper drainage is not included, which is a major concern in a city like Chicago, given its weather patterns. Figure 9 [9] illustrates an example of the draining techniques that can be used to prevent moisture buildup.

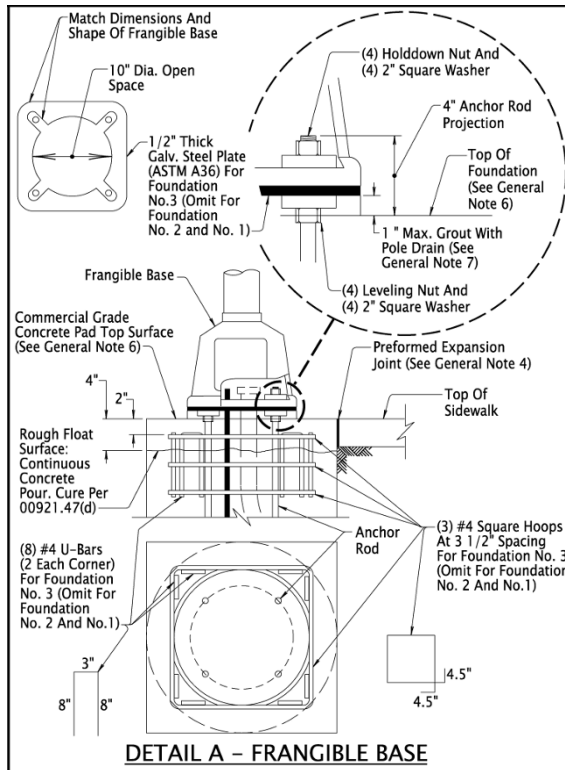


Figure 7: Gusseted Box Beam Arm Example



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Figure 8: Gusseted Box Beam Arm Example 2



General Notes:

1. All Bolts, Nuts And Washers To Conform To 02560.20 And Be Galvanized Steel According To 02560.40 Unless Noted Otherwise.
2. All Anchor Rods To Be Galvanized Steel Conforming To 02560.30.
3. All Pole Entrances Containing Wiring To Be Smooth.
4. Install 1/4" To 3/8" Thick Preformed Expansion Joint Filler Around Footing In Sidewalk Areas.
5. The Entire Foundation To Be Located On A Single Plane With Less Than 2% Slope. The Flat Edge(s) Of The Foundation May Be Adjacent To The Turn Space, Back Of Walk, Or A Curb Ramp Grade Break Line.
6. Install Commercial Grade Concrete Pad Above Rough Float Surface With Top Surface Matching Sidewalk Grade And Less Than 1/4" Vertical Exposure From Adjacent Grade. Clean Rough Float Surface Prior To Placing Fresh Concrete By Removing All Scum, Laitance, Loose Gravel, And Sediment. Pour During Sidewalk Installation After Installing Pipe And Appurtenances.
7. Non-Shrink High Early Strength Grout (Non-Ferrous) with 3/4" Diameter Pole Drain And A Minimum Strength of 5000 psi. Do Not Use Footing Concrete.

Figure 9: Pole Moisture Draining Method Example

Objectives

Category	Objectives
Structural Integrity	Stable, Durable,
Aesthetics	Rustic, Classic
Cost	Affordable, Efficient
Environmental Impact	Eco-friendly, Weather-resistant
Safety	Visible, Compliant
FEA Performance	Accurate, Validated, Precise

Initial Brainstorming

The sketch below is of the initial design. The idea is to combine the truss cantilever and decorative pole designs, and to add a swept or curved feature to the mast arm. The base of the light pole would have a flared or chamfered base in order to provide additional support.

Various loads will be tested in the FEA simulation. First, the effect of high winds will be tested via conducting a steady-state CFD simulation. A static load test will be performed as well to ensure the design can support the static loads due to the weight of the parts as well. For this simulation to be accurate, the base of the structure will be set as a fixed support, the various parts of the structure will be assigned appropriate materials, and additional pressure loads will be applied to simulate potential ice buildup.

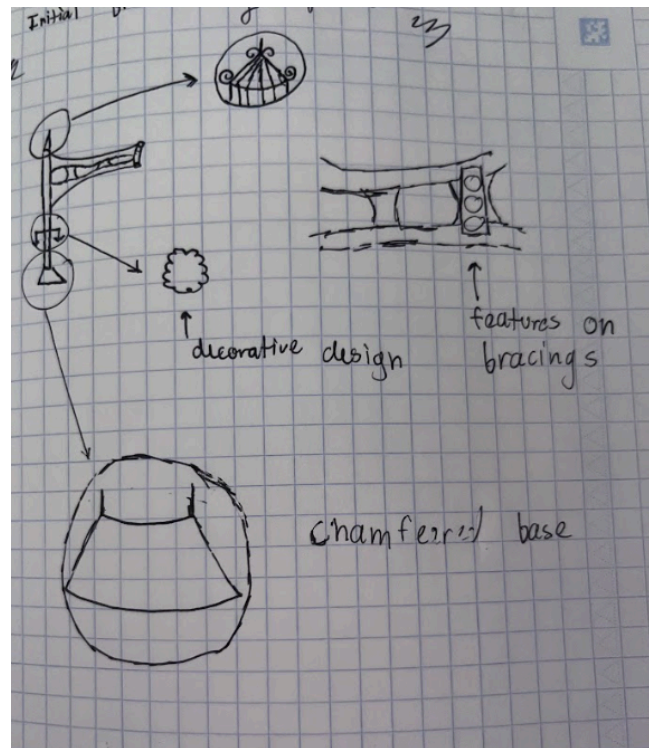


Figure 10: Initial Sketches

3. Design Process, FEA, and Evolution

This section details the process of FEA over the course of completing this project. Through many design iterations and peer reviews, a design that yielded favorable FEA results was created.

3a. Initial Concept

The original design included a truss system in the mast arm, a hexagonal cross-section of the vertical pole, and nonstructural, ornamental features. The original intent was to have a more rustic look, as shown below.

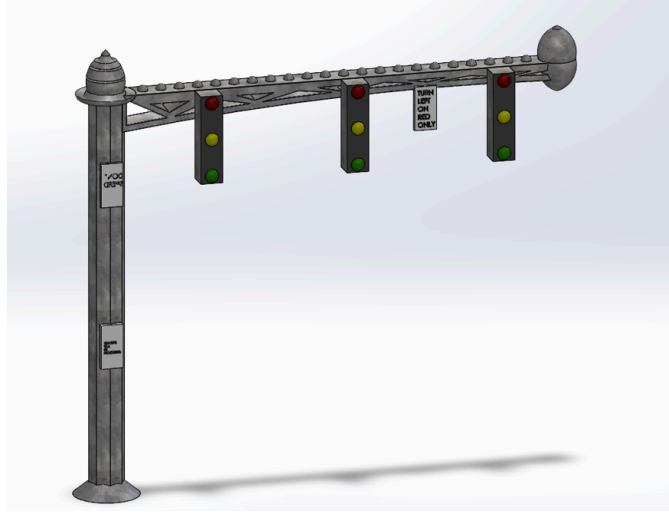


Figure 11: Original Mast Arm Design

3b. FEA

The FEA setup included a fixed support on the base of the vertical pole, standard earth gravity, two forces for each sign and light (one force for the weight, and one force for the wind), and a pressure representing the dynamic wind pressure applied to one side of the structure. The values for the different forces and pressures, and the formulas used to calculate the various forces and pressures.

Values For Each Force

Force	Formula	Values
Weight of Sign	Weight given by project rules	5 lbs
Wind Force on Sign	$F_{wind} = P_{dynamic} \cdot A_{front}$	208.8 N
Weight of Light	Weight given by project rules	60 lbs
Wind Force on Light	$F_{wind} = P_{dynamic} \cdot A_{front}$	416.56 N

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Dynamic Wind Pressure	$P_{dynamic} = \frac{1}{2} \rho v^2$	$v = 95.3 \text{ mph}, P_{dyn} = 1151 \text{ Pa}$
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The material of the structure is structural steel, and the mesh had the settings detailed in the image below. Note that the nodes and elements count is quite low, however, this is due to the fact that this project was completed on the free student version of Ansys, and so there is a cap to how many nodes and elements are allowed.

Physics Preference	Mechanical	Transition	Slow
Element Order	Program Controlled	Span Angle Center	Fine
☐ Element Size	Default	Initial Size Seed	Assembly
Sizing		Bounding Box Diagonal	9.6595 m
Use Adaptive Sizing	Yes	Average Surface Area	0.22513 m ²
Resolution	3	Minimum Edge Length	1.0639e-002 m
Mesh Defeaturing	Yes	Quality	
☐ Defeature Size	Default	Inflation	
Transition	Slow	Advanced	
Span Angle Center	Fine	Automatic Methods	
Initial Size Seed	Assembly	Sheet Body Method	Prime Quad Dominant
Bounding Box Diagonal	9.6595 m	Sweepable Body Method	Sweep
Average Surface Area	0.22513 m ²	Statistics	
Minimum Edge Length	1.0639e-002 m	☐ Nodes	60814
		☐ Elements	36169
		Show Detailed Statistics	No

Figure 12: Mesh Settings

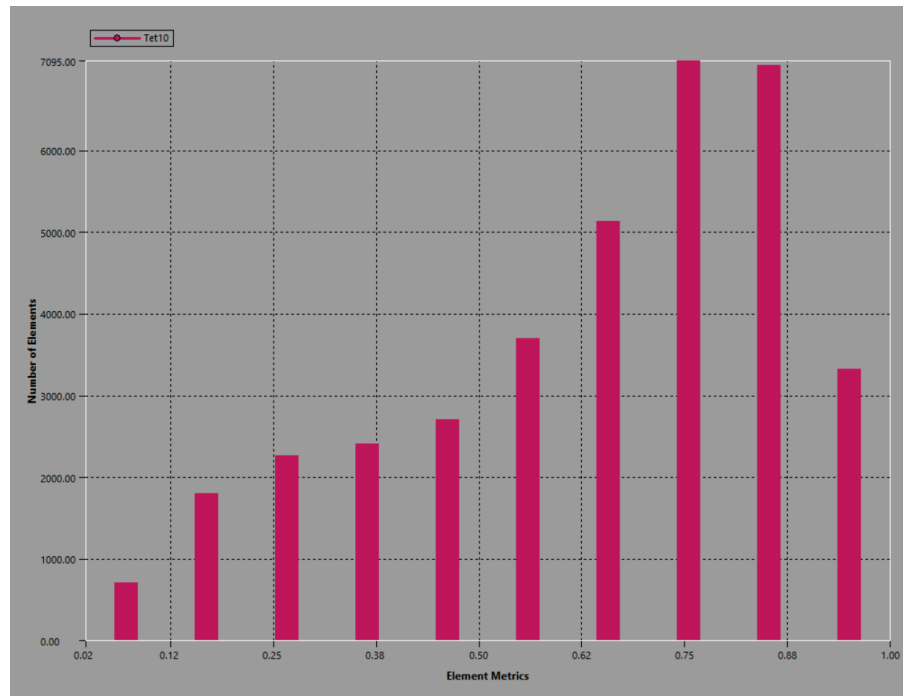


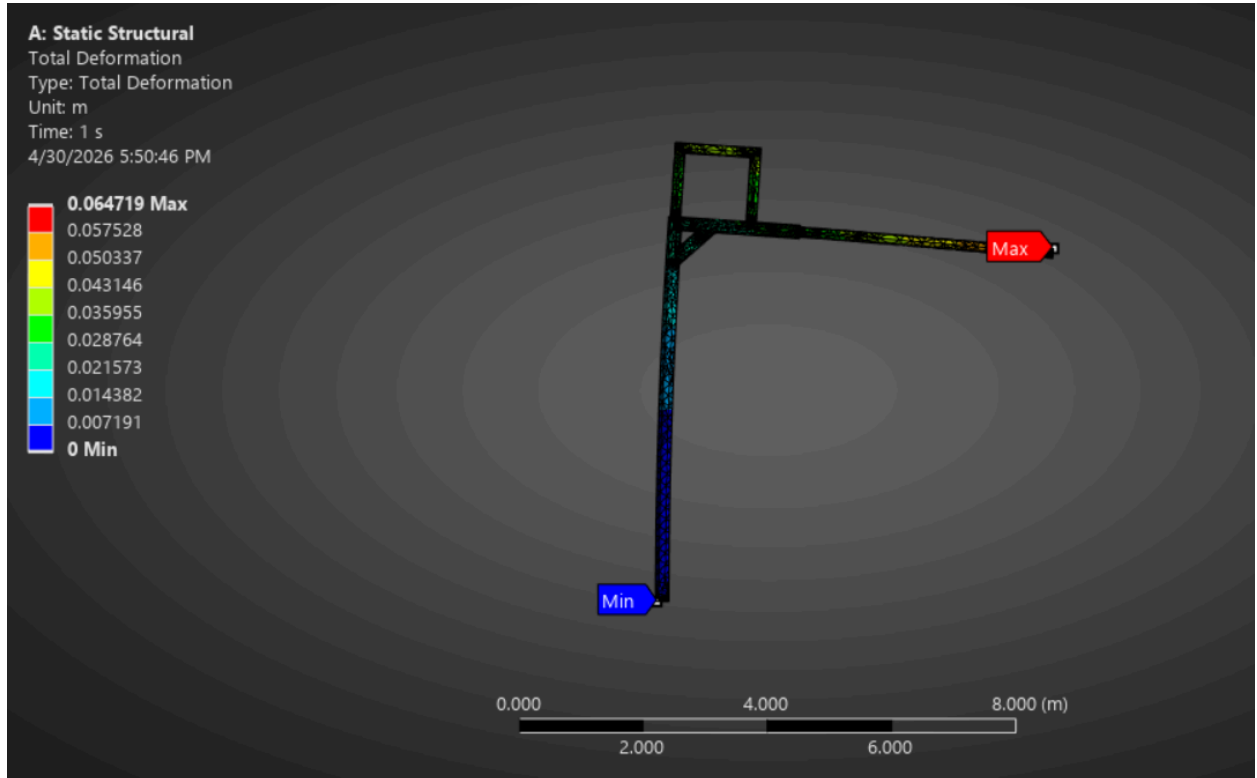
Figure 13: Mesh Metrics

The graph above shows that while the mesh element and node count is generally low, the metrics are still right-leaning, which is a good sign that the mesh will yield accurate results.

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The tests run by the simulation include total deformation, equivalent stress, and the safety factor stress tool. The results from each test are shown in the images below. Note that the FOS never dips below 2, and the maximum deformation does not exceed 0.067 meters. This max deformation cap was calculated in accordance with the project's rules. The length of the mast arm was divided by 100 mm in order to find that .067 value.



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Figure 14: Deformation Results

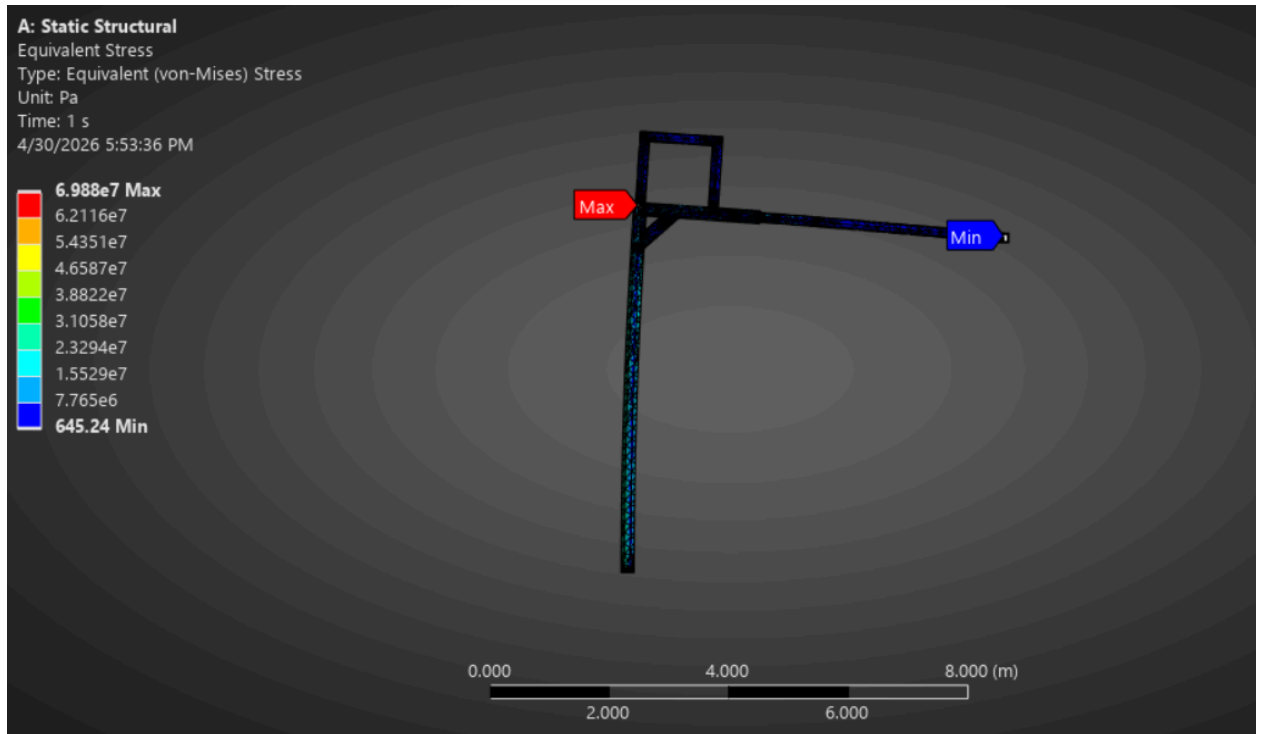


Figure 15: Equivalent Stress

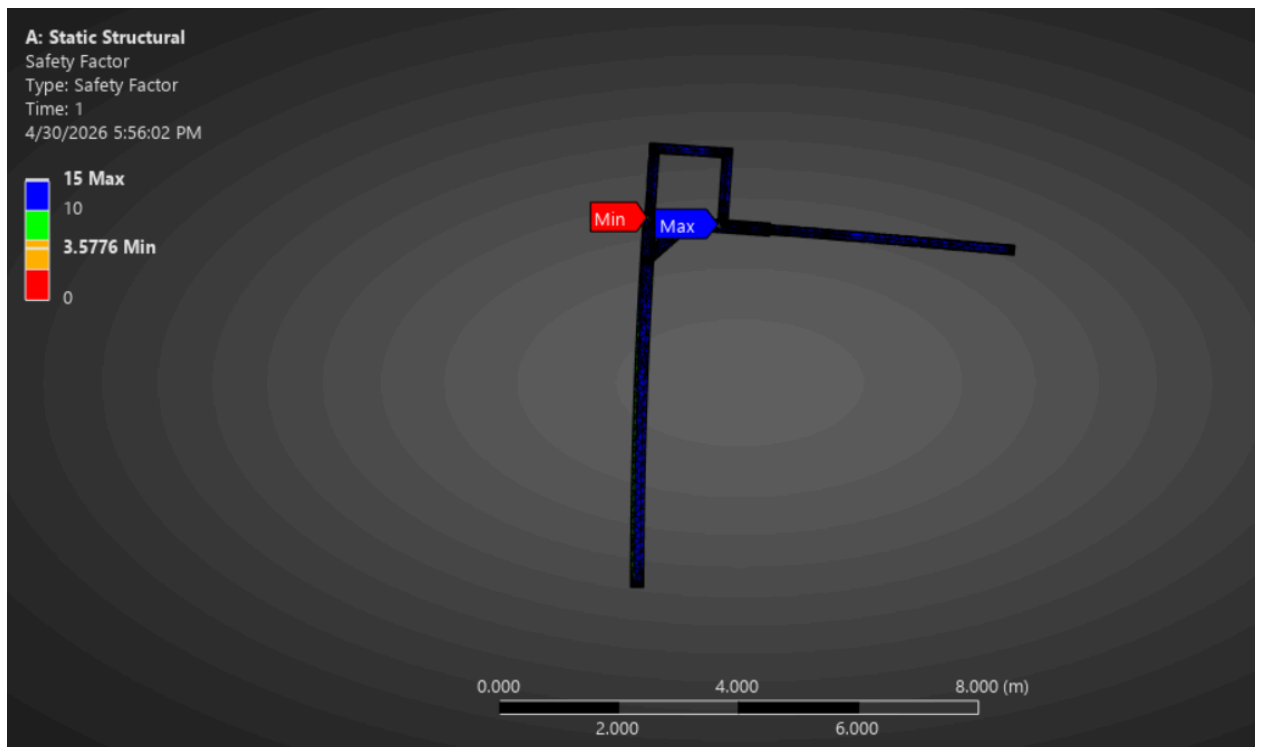


Figure 16: Factor of Safety (Stress Tool)

3c. Iterations and Improvements

The first iteration was the most complex. As the project developed, the design got simpler. Fewer trusses were incorporated after each iteration until there were no more triangles visible. The location of the hanging masses was more condensed as the iterations went on.

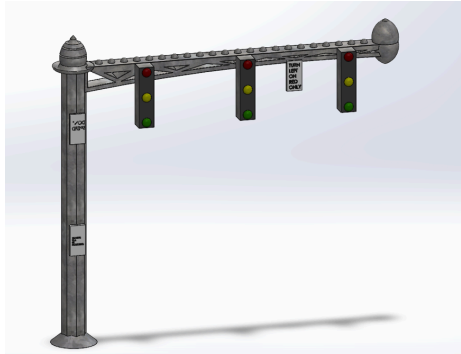


Figure 17: Iteration 1

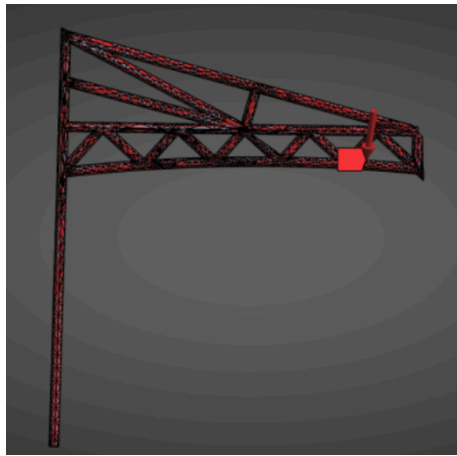
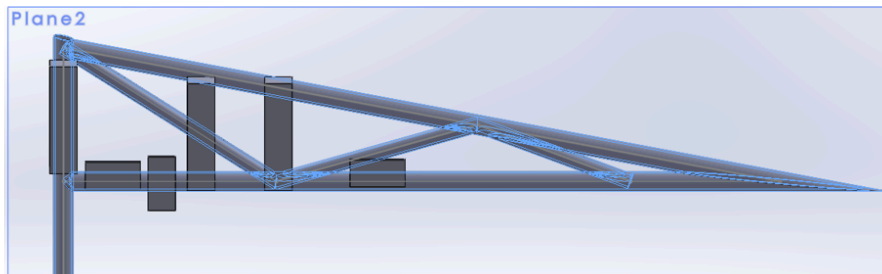


Figure 18: Iteration 2



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Figure 19: Iteration 3

After iteration 3, the triangular truss design was abandoned due to too low a FOS, and too much deformation. The locations of the hanging masses were also condensed in order to yield better FEA results.

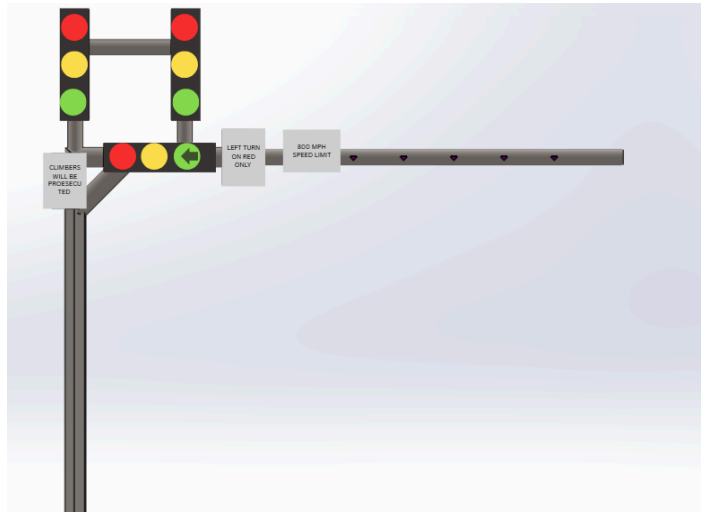


Figure 20: Iteration 4

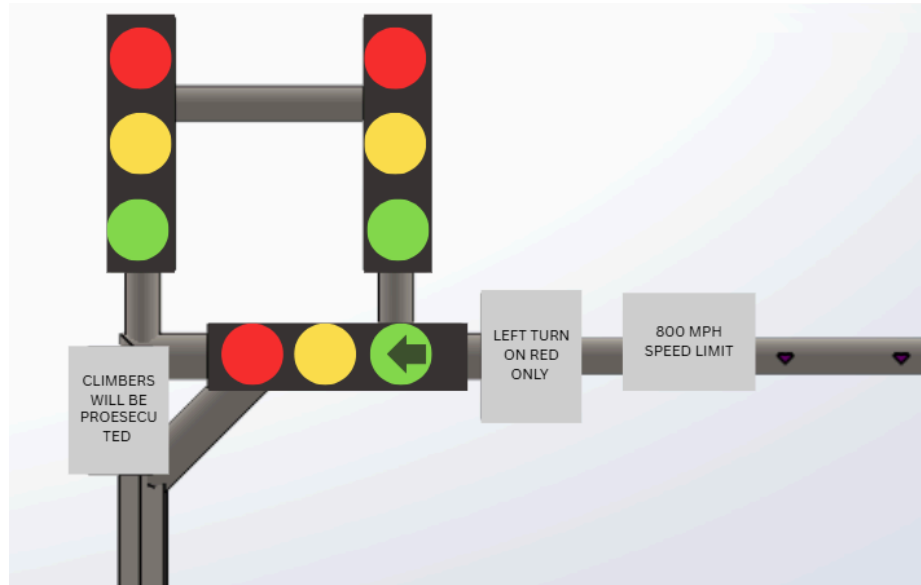


Figure 21: Close Up of Signs

After the 4th iteration, the simulations yielded favorable results, and so the aesthetics were given more attention. Beyond the structural components, pink hearts were added to the mast arm, and humorous signs were installed. The most important sign is the one that warns pedestrians that they will be prosecuted for climbing the traffic light. While there could have been some geometry

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added to physically limit people from climbing the structure, the designer determined that the most effective way to do so was to incorporate the threat of government prosecution.

3c. Challenges Faced

A variety of difficulties were encountered. First, there was a challenge with creating an aesthetically pleasing design that also passes the simulation. Prioritizing aesthetics before achieving successful simulation results created a very time-inefficient workflow, and the designer eventually realized that it is most efficient to first create a simple structure that passes the simulation and later add nonstructural, aesthetic features. The next problem was ensuring the FOS and deformation were within the project's rules. This was also addressed by prioritizing a stable structure over aesthetics at first. Another challenge faced was the time constraint. The time needed to iterate between SolidWorks and Ansys took more time than anticipated. This was addressed by spending less time on final aesthetic details and more time on the structural elements in order to get the simulation. The end of the project felt quite rushed as a result. Many meshing errors were encountered due to the node and element limits set by the student version of Ansys. This was addressed by lowering the resolution and size of the mesh, and by making the geometry simple enough so that a lower element and node count would not jeopardize the accuracy of the FEA results.

3d. Final Design

The final design of the mast arm is a simple structure with members resembling a cube, hence the name CUBE. The vertical pole is 24 feet tall, and the mast arm spans 20 feet, which appropriately covers the majority of a 26-foot-wide road. Figure 20 illustrates the final CAD model, along with added symbols and shapes representing the aesthetic features of the traffic lights and signs. Figures 14-16 illustrate the favorable results of the FEA simulations. Figure 21 illustrates a closer look at the signs and traffic lights incorporated into the design. Note that instead of creating complex geometry that would physically prevent revelers, a sign warning potential revelers of prosecution was incorporated into the design.

As shown in the figure below, the span of the mast arm is 20 feet, not 26 feet. While this mast arm does not span the full 26 feet, it is still long enough to be comfortably within the visual field of anyone on a 26-foot road under this structure, and the shorter length yielded better FEA results. The vertical pole is also taller than the minimum requirement specified by the project rules. This is because in order for the structure to be stable, the designer did not place any mass components below 18 feet, including any supports attached to the vertical pole. Note that the hearts are not dimensioned out, because they are not structural components.

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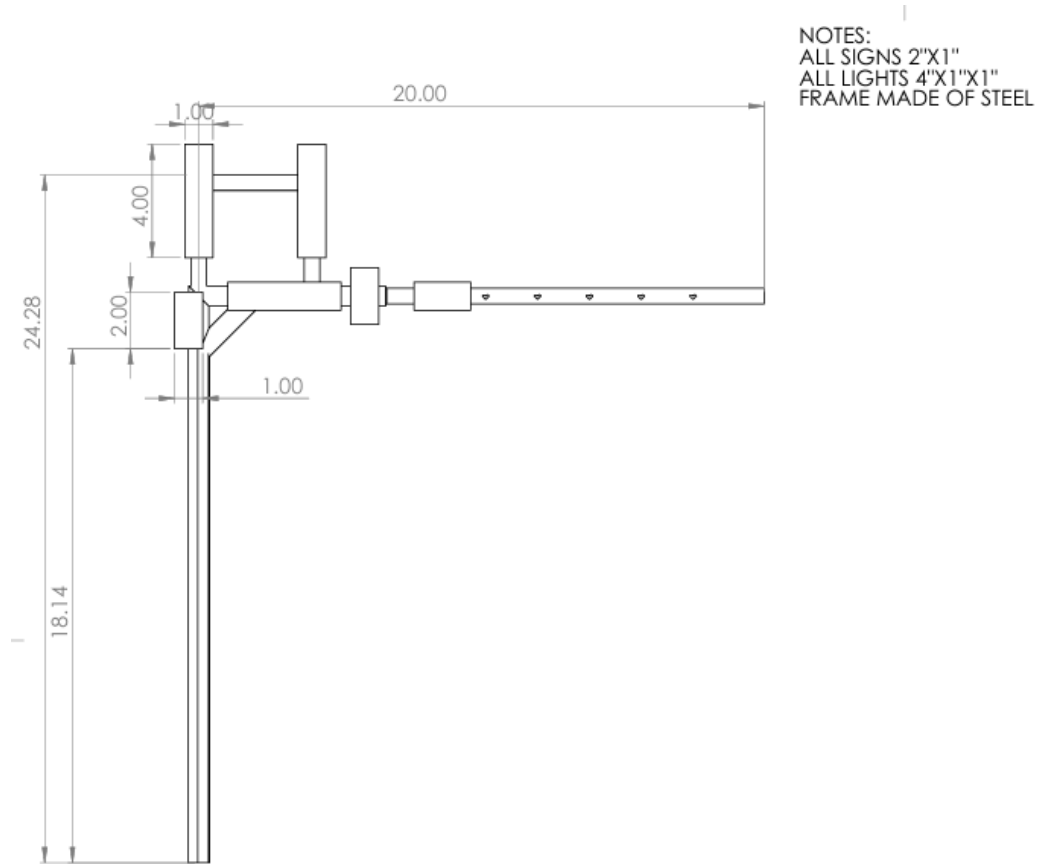


Figure 22: Dimensions of Structure, Final Design

4. Cost

The table below breaks down the cost of the structural elements. Note that the price estimation of the specified material[10] was researched prior to these calculations in order for an accurate cost to be estimated.

Cost Evaluation

Item	Material	Cost per kg	Mass(kg)	Total cost
Frame	Structural Steel	\$3.61	2145.17 kg	\$7740.71

5. Self-Reflection

After completing this project, many lessons were learned. Beyond learning more about Ansys software features, this project provided an opportunity to practice maximizing the efficiency of the design iteration process and problem-solving. The nature of this project requires a lot of iteration between SolidWorks and Ansys, and with each new software introduced to the process, new problems can arise. This project was challenging to the creative process because while it was easy enough to come up with interesting and unique designs, making those designs structurally sound was challenging. Because of this, there was often a trade-off between a creative or aesthetically pleasing design and a structurally sound design that met the project's requirements. This also created a new opportunity to problem solve in the sense that it was difficult to get agreeable results from Ansys while still maintaining some sort of creative design.

It is also worth noting that the time constraints of this project were particularly challenging due to responsibilities beyond ME 347. As an engineering student at UIC, I am also the Vice President of AIAA, and our competition took place right around the project 2 deadline. Due to this, I did not spend as much time as I should have on this project, and completing it on time was quite challenging. If more time were available, I would have spent more time adding non-structural, aesthetic details, and more time on making the structural elements more visually appealing or unique.

My final design is much simpler than the original one I had in mind. While it meets the project rules, the design is quite bare-bones aesthetically. More specifically, I wanted the mast arm to have a more unique design, but the deflection rule made it difficult to have a more complex design.

Overall, this project taught me a lot about the trade-offs that must take place in order for engineers to truly solve problems in an efficient, timely manner.

6. References

Using the [IEEE style](#), include any and all external resources (e.g., websites, patents, journals, books, etc.) you have looked up and referenced in this report. Each reference listed in this section of the report must be cited in-text.

- [1] [SDD 9e8 Type 9, 10, 12 and 13 Poles with Monotube Arms](#)
- [2] [Decorative Poles - Traffic Safety Supply Company](#)
- [3] [Decorative Aluminum Mast Arm](#)
- [4] [Sign Structures | Valmont Structures](#)
- [5] [Truss Arms - Hapco Pole Products](#)
- [6] [\(PDF\) Fatigue Life Evaluation of High-Mast Lighting Tower \(HMLT\) and Aluminum Poles for Traffic Signals and Luminaires](#)
- [7] [Beam - Outdoor Metal Light Pole - Structura](#)
- [8] [Professional Manufacture Outdoor Street Light Pole Single Double Arm Traffic Light Pole Galvanized Pole - Street Light Pole and Light Pole](#)
- [9] [2026 11x17 Class Materials](#)

